

Random slopes

Logistics

- + Project part 3 due last day of class
 - + No additional analysis needed; just combining work from first two parts and adding introduction, conclusion, etc.
 - + Work time in class on Wednesday and Friday
- + Homework 9 (optional) due Wednesday
- + Final exam: December 13, 2 - 5pm in Carswell 101 (our usual room)
 - + Final exam will be cumulative, and will include mixed effects models

Revisiting the shape assumption

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- + This model assumes that the slope is the same for each neighborhood

How can we change the model to allow the slope to be different in different neighborhoods?

Adding random slopes

$$Price_{ij} = \underbrace{\beta_0 + u_i}_{\text{intercept for neighborhood } i} + \underbrace{(\beta_1 + v_i)}_{\text{slope for neighborhood } i} Satisfaction_{ij} + \varepsilon_{ij}$$

- + β_0 = mean price when satisfaction is 0 (average across neighborhoods)
- + $\beta_0 + u_i$ = mean price when satisfaction is 0 in neighborhood i
- + β_1 = average change in price for a one-unit increase in satisfaction (average across neighborhoods)
- + $\beta_1 + v_i$ = average change in price for a one-unit increase in satisfaction in neighborhood i

Warmup

Work on the warmup activity (handout)

Warmup

The researchers hypothesize that anxiety levels depend on the type of performance (large or small ensembles), and that the difference in anxiety levels between large and small ensembles varies from person to person.

What mixed effects model should the researchers use to investigate their hypothesis?

$$\text{Anxiety}_{ij} = \beta_0 + u_i + (\beta_1 + v_i) \text{Large}_{ij} + \xi_{ij}$$

↑
anxiety of student
i before performance j

Example: music performance anxiety

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

Interpret the fixed effects and random effects in the model.

Example: music performance anxiety

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

Interpret the fixed effects and random effects in the model.

- + β_0 = average performance anxiety before small ensemble and solo performances (average across musicians)
- + $\beta_0 + u_i$ = average performance anxiety before small ensemble and solo performances for musician i
- + β_1 = average difference in anxiety before large ensemble performances (compared to small/solo performances) (average across musicians)
- + $\beta_1 + v_i$ = average difference in anxiety before large ensemble performances for musician i

Fitting a model with random slopes

```
m1 <- lmer(na ~ large + (large|id),  
           data = music)
```

- + This means we include `large` as a fixed effect, *and* we allow the coefficient on `large` to vary from individual to individual

Fitting a model with random slopes

```
m1 <- lmer(na ~ large + (large|id),  
           data = music)  
summary(m1)
```

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	16.7297	0.4908	34.09
large	-1.6762	0.5425	-3.09

$$\hat{\beta}_0 = 16.73 \quad \hat{\beta}_1 = -1.68$$

How would I interpret $\hat{\beta}_0$ and $\hat{\beta}_1$?

Fitting a model with random slopes

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

$$\hat{\beta}_0 = 16.73 \quad \hat{\beta}_1 = -1.68$$

- ✚ The average anxiety before a small/solo performance is 16.73
- ✚ On average, student anxiety decreases by 1.68 before a large performance (compared to a small/solo performance)

Suppose $\hat{u}_1 = 1$ $\hat{v}_1 = -0.3$

$\Rightarrow$ For musician 1

$$\hat{Anxiety}_{1j} = (16.73 + 1) + (-1.68 - 0.3)Large_{1j}$$

$$= 17.73 - 1.98 Large_{1j}$$

17.73 = musician 1's average anxiety before small performances

17.73 - 1.98 = musician 1's average anxiety before large performances

Fitting a model with random slopes

```
summary(m1)
```

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
id	(Intercept)	6.3330	2.5165	
	large	0.7429	0.8619	-0.76
Residual		21.7712	4.6660	

$\hat{\sigma}_v^2$

$\hat{\sigma}_u^2$

correlation
between
random effects
u & v

$\hat{\sigma}_\varepsilon^2$

What does this output tell us about the random effects and the noise?

Fitting a model with random slopes

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
id	(Intercept)	6.3330	2.5165	
	large	0.7429	0.8619	-0.76
Residual		21.7712	4.6660	

- + $\hat{\sigma}_u^2 = 6.333$ (variability in anxiety before small performances, between students)
- + $\hat{\sigma}_v^2 = 0.743$ (variability in difference in anxiety before large performances, between students)
- + $\hat{\sigma}_\varepsilon^2 = 21.77$ (variability in anxiety between performances, within a student)

Correlation between slopes and intercepts

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
id	(Intercept)	6.3330	2.5165	
	large	0.7429	0.8619	-0.76
Residual		21.7712	4.6660	

- + $\hat{\rho}_{uv} = -0.76$ (estimated correlation between the random slope and random intercept for an individual)
- + Subjects with higher performance anxiety scores for solos and small ensembles tend to have greater decreases in performance anxiety for large ensemble performances

$\hat{\rho}_{uv} < 0 \Rightarrow$ as $u \uparrow$, $v \downarrow$ (on average)
higher anxiety in small performances
 $\Rightarrow$ greater decrease in anxiety for large performances

Writing down the model

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2) \quad \begin{bmatrix} u_i \\ v_i \end{bmatrix} \stackrel{iid}{\sim} N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_u^2 & \rho_{uv}\sigma_u\sigma_v \\ \rho_{uv}\sigma_u\sigma_v & \sigma_v^2 \end{bmatrix} \right)$$

Anybody know the name of this new thing?

Writing down the model

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

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Anybody know the name of this new thing?

A bivariate normal distribution!

Writing down the model

$$Anxiety_{ij} = \beta_0 + u_i + (\beta_1 + v_i)LargeEnsemble_{ij} + \varepsilon_{ij}$$

$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2) \quad \begin{bmatrix} u_i \\ v_i \end{bmatrix} \stackrel{iid}{\sim} N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_u^2 & \rho_{uv}\sigma_u\sigma_v \\ \rho_{uv}\sigma_u\sigma_v & \sigma_v^2 \end{bmatrix} \right)$$

- + This just says that both u_i and v_i come from a normal distribution
 - + the variance of u_i is σ_u^2
 - + the variance of v_i is σ_v^2
 - + the correlation between u_i and v_i is ρ_{uv}
- + *Note:* the population model includes the distribution of the random effects and noise

Class activity

Work on the class activity on the course website. Submit your rendered HTML on Canvas at the end of class.